I. TERMS PROPORTIONAL TO 1

A. Four ρ contributions

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\\
&\Bigg\{-\left[U^{ab'}(x')\rho^{b'}(x')-\rho^{b'}(x')U^{ab'}(w')\right]\times\left[U^{ab}(u)\rho^b(u)\rho^c(x)\rho^c(y)-U^{ab}(w)U^{cd}(x)\rho^d(x)U^{ce}(y)\rho^e(y)\rho^b(u)\right]\\
&+\left[U^{ab'}(x')\rho^{b'}(x')-\rho^{b'}(x')U^{ab'}(w')\right]\times\left[U^{ab}(w)\rho^c(x)\rho^c(y)\rho^b(u)-U^{cd}(x)\rho^d(x)U^{ce}(y)\rho^e(y)U^{ab}(u)\rho^b(u)\right]\\
&-\left[U^{ab'}(x')\rho^{b'}(x')-\rho^{b'}(x')U^{ab'}(w')\right]\times\Big[U^{ac}(w)U^{bd}(z)U^{be}(y)\rho^e(y)\{\rho^d(x),\rho^c(u)\}\\
&-U^{bc}(y)\rho^c(y)\Big(U^{bd}(x)\rho^d(x)U^{ae}(u)\rho^e(u)+U^{ae}(u)\rho^e(u)U^{bd}(x)\rho^d(x)\Big)\Big]\\
&-\left[U^{ab'}(x')\rho^{b'}(x')-\rho^{b'}(x')U^{ab'}(w')\right]\times\Big[\Big(U^{bc}(x)\rho^c(x)U^{ad}(u)\rho^d(u)+U^{ad}(u)\rho^d(u)U^{bc}(x)\rho^c(x)\Big)U^{be}(y)\rho^e(y)\\
&-\Big(U^{bc}(x)\rho^c(x)U^{ad}(u)\rho^d(u)+U^{ad}(u)\rho^d(u)U^{bc}(x)\rho^c(x)\Big)U^{be}(z)\rho^e(y)\Big]\Bigg\}+\text{c.c}\\
&+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\\
&\Bigg\{\frac{1}{2}\Big[\{\rho^b(x),\rho^{c'}(x')\}U^{ac'}(w')-\Big(U^{c'd'}(x)\rho^{d'}(x)U^{ad}(x')\rho^d(x')+U^{ad}(x')\rho^d(x')U^{c'd'}(x)\rho^{d'}(x)\Big)U^{c'b}(z)\Big]\\
&\times\Big[U^{ac}(w)\{\rho^c(u),\rho^b(y)\}-U^{cb}(z)\Big(U^{ce}(y)\rho^e(y)U^{ab'}(u)\rho^{b'}(u)+U^{ab'}(u)\rho^{b'}(u)U^{ce}(y)\rho^e(y)\Big)\Big]\\
&+2\Big[U^{ab'}(x')\rho^{b'}(x')U^{c'a'}(x)\rho^{a'}(x)U^{c'b}(z)-\rho^{c'}(x')U^{d'b}(z)U^{d'a'}(x)\rho^{a'}(x)U^{ac'}(w')\Big]\times\\
&\Big[U^{cb}(z)U^{cd}(y)\rho^d(y)U^{ae}(u)\rho^e(u)-U^{ac}(w)U^{de}(y)\rho^e(y)U^{db}(z)\rho^c(u)\Big]\\
&+2\Big[U^{c'd'}(x)\rho^{d'}(x)U^{ae'}(x')\rho^{e'}(x')U^{c'b}(z)-\rho^b(x)U^{ab'}(x')\rho^{b'}(x')\Big]\times\\
&\Big[U^{cb}(z)U^{ad}(u)\rho^d(u)U^{ce}(y)\rho^e(y)-U^{ac}(u)\rho^c(u)\rho^b(y)\Big]\Bigg\}\\
&+\Bigg\{\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\\
&\Bigg\{\Big[U^{ab'}(x')\rho^{b'}(x')U^{c'a'}(x)\rho^{a'}(x)U^{c'b}(z)-\rho^{c'}(x')U^{d'b}(z)U^{d'a'}(x)\rho^{a'}(x)U^{ac'}(w')\Big]\times\\
&\Big[U^{ac}(w)\{\rho^c(u),\rho^b(y)\}-U^{cb}(z)\Big(U^{ce}(y)\rho^e(y)U^{ae'}(u)\rho^{e'}(u)+U^{ae'}(u)\rho^{e'}(u)U^{ce}(y)\rho^e(y)\Big)\Big]\\
&+\Big[U^{c'd'}(x)\rho^{d'}(x)U^{ae'}(x')\rho^{e'}(x')U^{c'b}(z)-\rho^b(x)U^{ab'}(x')\rho^{b'}(x')\Big]\times\\
&\Big[U^{ac}(w)\{\rho^c(u),\rho^b(y)\}-U^{cb}(z)\Big(U^{ce}(y)\rho^e(y)U^{aa'}(u)\rho^{a'}(u)+U^{aa'}(u)\rho^{a'}(u)U^{ce}(y)\rho^e(y)\Big)\Big]\\
&+2\Big[U^{c'd'}(x)\rho^{d'}(x)U^{ae'}(x')\rho^{e'}(x')U^{c'b}(z)-\rho^b(x)U^{ab'}(x')\rho^{b'}(x')\Big]\times\\
&\Big[U^{cb}(z)U^{cd}(y)\rho^d(y)U^{ae}(u)\rho^e(u)-U^{ac}(w)U^{de}(y)\rho^e(y)U^{db}(z)\rho^c(u)\Big]\Bigg\}+\text{c.c}\Bigg\}
\end{aligned}
\tag{I.1}
$$

eq:4rho

Corrections (orientation of the z- and w'-Wilson lines in the second group). The eight Wilson lines marked in red above have been transposed so that the second group follows the same orientation convention as the third group: every $U(z)$ carries the cross-contracted index second (so that same-point contractions give $U^{d'b}(z)U^{cb}(z) = \delta^{d'c}$), and every $U(w')$ carries the external index a first. **Verified** (symbolically, arbitrary N_c): with these transpositions – and only with them, up to simultaneously transposing both members of a contracted pair – the fully symmetrised four-ρ sector of this equation cancels identically, consistent with the two- and four-Wilson-line sums displayed below. **Caveat:** the three-ρ and two-ρ reordering remainders quoted in the following subsections do not yet follow mechanically from this master equation even after the orientation fix, so its transcription from Ω should be revisited; those remainders themselves are certified independently (the collected three-ρ set cancels exactly, and the two-ρ $\log(k^+/\Lambda)$ set reproduces $-H_{\rm JIMWLK}\, dN_{\rm LO}$ exactly), and they, not this master equation, are what enters the evolution files. The terms with two Wilson lines are as follows

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right] + \log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\rho^{b'}(x')\rho^{b}(u)\rho^{c}($$
$$\Big[-U^{ab'}(x')U^{ab}(u) + U^{ab'}(w')U^{ab}(u) + U^{ab}(w)U^{ab'}(x') - U^{ab}(u)U^{ab'}(x') + U^{ab'}(x')U^{ab}(w) - U^{ab'}(w')U^{ab}(w)$$
$$+ U^{ab}(u)U^{ab'}(w') - U^{ab}(w)U^{ab'}(w') + 2U^{ab}(w)U^{ab'}(w') + 2U^{ab'}(x')U^{ab}(u)$$
$$- 2U^{ab'}(x')U^{ab}(w) - 2U^{ab}(u)U^{ab'}(w')\Big] = 0 \tag{I.2}$$

Similarly one can show that the contribution proportional to four Wilson lines identically vanishes. Terms with four

Wilson lines are as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \Bigg\{ \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\,\rho^{b'}(x')\rho^d(x)\rho$$

$$\Big[2\,U^{ab'}(x')U^{ac}(w)U^{bd}(x)U^{be}(y) - 2\,U^{ab'}(w')U^{ac}(w)U^{bd}(x)U^{be}(y) - 2\,U^{ab'}(x')U^{bd}(x)U^{be}(y)U^{ac}(u)$$
$$+ 2\,U^{ab'}(w')U^{bd}(x)U^{be}(y)U^{ac}(u) - 2\,U^{ab'}(x')U^{ac}(w)U^{bd}(z)U^{be}(y) + 2\,U^{ab'}(w')U^{ac}(w)U^{bd}(z)U^{be}(y)$$
$$+ 2\,U^{ab'}(x')U^{be}(y)U^{bd}(x)U^{ac}(u) - 2\,U^{ab'}(x')U^{bd}(x)U^{ac}(u)U^{be}(y) + 2\,U^{ab'}(x')U^{bd}(x)U^{ac}(u)U^{be}(z)$$
$$- 2\,U^{ab'}(w')U^{ac}(u)U^{bd}(x)U^{be}(z)\Big]$$
$$+ \Big[- 2\,U^{ac}(u)U^{ab'}(w')U^{bd}(z)U^{be}(y) + 2\,U^{ac}(w)U^{ab'}(w')U^{bd}(z)U^{be}(y) - 2\,U^{ac}(w)U^{be}(y)U^{bd}(x)U^{ab'}(x')$$
$$+ 2\,U^{ac}(u)U^{bd}(x)U^{ab'}(x')U^{be}(z) + 2\,U^{ac}(w)U^{bd}(x)U^{ab'}(x')U^{be}(y) - 2\,U^{ac}(w)U^{bd}(x)U^{ab'}(x')U^{be}(z)\Big]\Bigg\}$$
$$+ \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\,\rho^{b'}(x')\rho^d(x)\rho^c($$
$$\Big[- 2\,U^{ab'}(w')U^{bd}(z)U^{be}(y)U^{ac}(u) - 2\,U^{bd}(x)U^{ab'}(x')U^{be}(z)U^{ac}(w) + 2\,U^{bd}(x)U^{ab'}(x')U^{be}(y)U^{ac}(u)$$
$$+ 2\,U^{ab'}(x')U^{bd}(x)U^{be}(y)U^{ac}(u) - 2\,U^{ab'}(x')U^{bd}(x)U^{ac}(w)U^{be}(y) - 2\,U^{bd}(x)U^{ab'}(w')U^{be}(y)U^{ac}(u)$$
$$+ 2\,U^{bd}(x)U^{ab'}(w')U^{ac}(w)U^{be}(y) + 2\,U^{bd}(x)U^{ab'}(x')U^{ac}(u)U^{be}(y) - 2\,U^{bd}(x)U^{ab'}(x')U^{be}(z)U^{ac}(u)$$
$$- 2\,U^{ab'}(x')U^{bd}(z)U^{ac}(u)U^{be}(y)\Big]$$
$$+ \Bigg\{ \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\rho^{b'}(x')\,\rho^d(x)\rho$$

$$\Big[2\,U^{ab'}(x')U^{bd}(x)U^{be}(z)U^{ac}(w) - 2\,U^{ab'}(x')U^{bd}(x)U^{be}(y)U^{ac}(u) - 2\,U^{be}(z)U^{bd}(x)U^{ab'}(w')U^{ac}(w)$$
$$+ 2\,U^{bd}(x)U^{ab'}(w')U^{be}(y)U^{ac}(u) + 2\,U^{bd}(x)U^{ab'}(x')U^{be}(z)U^{ac}(w) - 2\,U^{bd}(x)U^{ab'}(x')U^{be}(y)U^{ac}(u)$$
$$+ 2\,U^{ab'}(x')U^{bd}(z)U^{be}(y)U^{ac}(u) + 2\,U^{bd}(x)U^{ab'}(x')U^{be}(y)U^{ac}(u) - 2\,U^{bd}(x)U^{ab'}(x')U^{ac}(w)U^{be}(y)$$
$$- 2\,U^{ab'}(x')U^{bd}(z)U^{be}(y)U^{ac}(u) + 2\,U^{ab'}(x')U^{ac}(w)U^{be}(y)U^{bd}(z)\Big]$$
$$+ \Big[2\,U^{ac}(u)U^{bd}(x)U^{be}(z)U^{ab'}(w') - 2\,U^{ac}(u)U^{bd}(x)U^{be}(y)U^{ab'}(x') - 2\,U^{be}(z)U^{bd}(x)U^{ac}(w)U^{ab'}(w')$$
$$+ 2\,U^{bd}(x)U^{ac}(w)U^{be}(y)U^{ab'}(x') + 2\,U^{bd}(x)U^{ac}(u)U^{be}(z)U^{ab'}(w') - 2\,U^{bd}(x)U^{ac}(u)U^{be}(y)U^{ab'}(x')$$
$$+ 2\,U^{ac}(u)U^{bd}(z)U^{be}(y)U^{ab'}(x') + 2\,U^{bd}(x)U^{ac}(u)U^{be}(y)U^{ab'}(x') - 2\,U^{bd}(x)U^{ac}(u)U^{ab'}(w')U^{be}(y)$$
$$- 2\,U^{ac}(u)U^{bd}(z)U^{be}(y)U^{ab'}(x') + 2\,U^{ac}(u)U^{ab'}(w')U^{be}(y)U^{bd}(z)\Big]\Bigg\} \tag{I.3}$$

Simplifying the terms with four Wilson lines, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \Bigg\{ \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,u,x',w,w'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(u-w)}{(x'-w')^2(u-w)^2}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\rho^{b'}(x')\rho^d(x$$

$$\Big[4\,U^{ab'}(w')U^{ac}(w)U^{bd}(z)U^{be}(y) + 2\,U^{ab'}(x')U^{ac}(u)U^{bd}(x)U^{be}(z) - 4\,U^{bd}(x)U^{be}(z)U^{ab'}(w')U^{ac}(w)$$
$$- 2\,U^{ab'}(x')U^{ac}(u)U^{bd}(z)U^{be}(y) - 2\,U^{ab'}(w')U^{ac}(u)U^{bd}(z)U^{be}(y) + 2\,U^{ab'}(w')U^{ac}(u)U^{bd}(x)U^{be}(z)\Big] \tag{I.4}$$

Since this equation is symmetric when $x \leftrightarrow y$ and by doing the algebra explicitly, we get all the terms that corresponds to four Wilson lines and ρ's to be 0.

B. Three ρ contributions

The three ρ contribution would arise from reordering the four ρ contributions. For our convenience we will write it as three ρ contribution with two, three and four Wilson lines.

1. *Two Wilson lines*

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\ \frac{(-w'+x')\cdot(-w+y)(x-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(x-z)^2(y-z)^2} f^{abc}\rho^d(x')\rho^a(x)\rho^c(y)\Big[U^{ed}(w')U^{eb}(w)-U^{eb}(w)U^{ed}(x')\Big] \\ +\frac{(-w'+x')\cdot(-w+y)(x-z)\cdot(x'-z)}{(-w'+x')^2(-w+y)^2(x-z)^2(x'-z)^2} f^{abc}\rho^c(x')\rho^a(x)\rho^d(y)\Big[U^{eb}(w')U^{ed}(y)-U^{eb}(w')U^{ed}(w)\Big]\Bigg] \tag{I.5}$$

We write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\Big[U^{ed}(w')U^{eb}(w)-U^{eb}(w)U^{ed}(x')\Big] \\ +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^c(x')\rho^a(y)\rho^d(x)\Big[U^{eb}(w')U^{ed}(x)-U^{eb}(w')U^{ed}(w)\Big]\Bigg] \tag{I.6}$$

We rewrite them (by some simplification)

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{ed}(w')U^{eb}(w) \\ -\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{de}(x') \\ +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{de}(x) \\ -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{db}(w')U^{de}(w)\Bigg] \tag{I.7}$$

2. *Three Wilson lines*

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\ -\frac{(-w'+x')\cdot(-w+y)(x-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(x-z)^2(y-z)^2} f^{abc}\rho^d(x')\rho^f(x)\rho^e(y)U^{bd}(w')U^{ce}(y)U^{af}(x) \\ +\frac{(-w'+x')\cdot(-w+x)(y-z)\cdot(x'-z)}{(-w+x)^2(-w'+x')^2(x'-z)^2(y-z)^2} f^{abc}\rho^e(x')\rho^d(x)\rho^f(y)U^{ce}(x')U^{bd}(w)U^{af}(y) \\ -\frac{(-w'+x')\cdot(-w+x)(y-z)\cdot(x'-z)}{(-w+x)^2(-w'+x')^2(x'-z)^2(y-z)^2} f^{abc}\rho^e(x')\rho^d(x)\rho^f(y)U^{ce}(x')U^{bd}(x)U^{af}(y) \\ +\frac{(-w'+x')\cdot(-w+y)(x-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(x-z)^2(y-z)^2} f^{abc}\rho^d(x')\rho^f(x)\rho^e(y)U^{ce}(y)U^{bd}(x')U^{af}(x)\Bigg] \tag{I.8}$$

We write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^f(y)\rho^e(x)U^{bd}(w')U^{ce}(x)U^{af}(y)$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^d(x)\rho^f(y)U^{ce}(x')U^{bd}(w)U^{af}(y)$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^d(x)\rho^f(y)U^{ce}(x')U^{bd}(x)U^{af}(y)$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^f(y)\rho^e(x)U^{ce}(x)U^{bd}(x')U^{af}(y)\Bigg] \tag{I.9}$$

Rewriting with some simplification, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{bd}(w')U^{ce}(x)U^{ah}(y)$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x')U^{be}(w)U^{ah}(y)$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x')U^{be}(x)U^{ah}(y)$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{cd}(x)U^{be}(x')U^{ah}(y)\Bigg] \tag{I.10}$$

3. *Four Wilson lines*

$$\frac{d^3N_{\rm NLO}}{d^3k} = 0 \tag{I.11}$$

C. Two ρ contribution

The two ρ contribution arises from reordering four ρ' s and three ρ's. We write it in two parts:

1. *First part*

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{N_c}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$+2\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(x'-z)}{(-w'+x')^2(-w+y)^2(x'-z)^4}\,\rho^a(x')\rho^b(y)U^{ca}(x')U^{cb}(w)$$
$$-2\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(x'-z)}{(-w'+x')^2(-w+y)^2(x'-z)^4}\,\rho^a(x')\rho^b(y)U^{ca}(x')U^{cb}(y)$$
$$+2\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}\,\rho^b(x')\rho^a(y)U^{ca}(y)U^{cb}(w')$$
$$-2\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}\,\rho^b(x')\rho^a(y)U^{ca}(y)U^{cb}(x')$$
$$+3\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{2(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}\,\rho^b(x')\rho^a(y)U^{cb}(x')U^{ca}(y)$$
$$-8\frac{(-w'+x')\cdot(-w+x')(y-z)\cdot(x'-z)}{(-w+x')^2(-w'+x')^2(x'-z)^2(y-z)^2}\,\rho^b(x')\rho^a(y)U^{cb}(x')U^{ca}(y)\Bigg] \tag{I.12}$$

2. Second part

$$
\begin{aligned}
\frac{d^3N^{(1)}_{\rm NLO}}{d^3k} = {}& \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&-\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{def}\rho^{b'}(x')\rho^c(y)U^{eb}(w)U^{fb'}(x')U^{da}(y) \\
&+3\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{deb'}\rho^{b'}(x')\rho^c(y)\,U^{ab'}(y)U^{db'}(w')U^{eb}(z) \\
&+\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{2(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{ade}\rho^e(x')\rho^c(y)U^{b'd}(w')U^{b'b}(w) \\
&-\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{ade}\rho^e(x')\rho^c(y)U^{b'b}(w)U^{b'd}(x') \\
&+2\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{ade}\rho^e(x')\rho^c(y)U^{b'b}(y)U^{b'd}(x') \\
&+\frac{(-w'+y)\cdot(-w+x')(x'-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{dae}\rho^e(x')\rho^c(y)U^{b'b}(w')U^{b'd}(x') \\
&-\frac{3(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fa}(z)U^{d'b}(w)U^{fd}(y)U^{d'b'}(w') \\
&+2\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fa}(z)U^{d'b}(y)U^{fd}(y)U^{d'b'}(w') \\
&-3\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fa}(w)U^{d'b}(y)U^{d'd}(z)U^{fb'}(w') \\
&+3\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{d'a}(z)U^{fd}(w)U^{fb'}(w')U^{d'b}(y) \\
&+\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fd}(w)U^{fb'}(x')U^{d'a}(z)U^{d'b}(y) \\
&+\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{2(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{d'fl}\rho^{b'}(x')\rho^c(y)U^{d'a}(z)U^{fb}(w)U^{lb'}(x') \\
&+\frac{(-w'+x')\cdot(-w+y)(x'-z)\cdot(y-z)}{2(-w'+x')^2(-w+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{d'fl}\rho^c(x')\rho^{b'}(y)U^{d'a}(z)U^{fb}(w')U^{lb'}(y) \\
&-\frac{(-w'+y)\cdot(-w+x')(x'-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(x'-z)^2(y-z)^2}f^{abc}f^{d'fl}\rho^{b'}(x')\rho^c(y)U^{fb}(y)U^{lb'}(x')U^{d'a}(z) \\
&-\frac{(-w'+y)\cdot(-w+x')(y-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fb}(y)U^{fb'}(w)U^{d'a}(z)U^{d'd}(y) \\
&+\frac{(-w'+y)\cdot(-w+x')(y-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fb}(y)U^{fb'}(x')U^{d'a}(z)U^{d'd}(y) \\
&+2\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{d'a}(z)U^{fb}(y)U^{d'd}(y)U^{fb'}(x') \\
&+\frac{(-w'+y)\cdot(-w+x')(y-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fd}(y)U^{fb'}(w)U^{d'a}(z)U^{d'b}(y) \\
&-2\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fd}(y)U^{fb'}(w')U^{d'a}(z)U^{d'b}(y) \\
&-2\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fd}(y)U^{fb'}(x')U^{d'a}(z)U^{d'b}(y) \\
&-\frac{(-w'+y)\cdot(-w+x')(y-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(y-z)^4}f^{abc}f^{cde}\rho^{b'}(x')\rho^e(y)U^{fd}(y)U^{fb'}(x')U^{d'a}(z)U^{d'b}(y)\Bigg]
\end{aligned}
\tag{I.13}
$$

$$\frac{d^3N^{(2)}_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4} f^{abc}f^{cde}\rho^{b'}(x')\rho^{e}(y)U^{d'a}(w)U^{d'b'}(x')U^{fb}(z)U^{fd}(y)$$
$$+\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4} f^{abc}f^{cde}\rho^{b'}(x')\rho^{e}(y)U^{d'b}(w)U^{d'b'}(x')U^{fa}(y)U^{fd}(z)$$
$$-\frac{(-w'+y)\cdot(-w+x')(x'-z)\cdot(y-z)}{(-w+x')^2(-w'+y)^2(x'-z)^2(y-z)^2} f^{fd'l}f^{deb'}\rho^{b'}(x')\rho^{c}(y)U^{fd}(z)U^{lc}(y)U^{d'e}(x')$$
$$+\frac{(-w'+x')\cdot(-w+y)(y-z)\cdot(y-z)}{(-w'+x')^2(-w+y)^2(y-z)^4} f^{abc}f^{deb'}\rho^{f}(x')\rho^{c}(y)U^{da}(z)U^{b'b}(y)U^{ef}(x')\Bigg] \tag{I.14}$$